

L17



L17: Primitive Roots for prime numbers

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Theorem 5.9 (Legendre): Let p be a prime number and $d \in \mathbb{Z}$ with $d > 0$ and $d \mid p-1$.

Then there are exactly $\phi(d)$ incongruent integers having order d modulo p .

Proof: Given a positive divisor d of $p-1$, let $f(d)$ be the number of integers between 1 and $p-1$ inclusive having order d . We must show that $f(d) = \phi(d)$ for all such d . By Proposition 5.1 we know that each integer between 1 and $p-1$ inclusive has order dividing $p-1$. Thus

$$\sum_{d \mid p-1, d > 0} f(d) = p-1.$$

By Theorem 3.5, we have that

$$\sum_{d \mid p-1, d > 0} \phi(d) = p-1.$$

Therefore

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$$\sum_{d|p-1, d>0} f(d) = \sum_{d|p-1, d>0} \phi(d). \quad ①$$

In view of ①, to show $f(d) = \phi(d)$ for all positive divisors d of $p-1$, it suffices to show that $f(d) \leq \phi(d)$ for all such d .

If $f(d) = 0$, then $f(d) < \phi(d)$ is obvious.

Assume that $f(d) > 0$. Then there is an integer between 1 and $p-1$ inclusive, say a , such that $\text{ord}_p a = d$. The d integers $a, a^2, \dots, a^d$ are incongruent modulo p since $\text{ord}_m a = d$ (otherwise $a^i \equiv a^j \pmod{p}$ for

some $i > j$, and after applying the inverse of $a^j \pmod{p}$ we have $a^{i-j} \equiv 1 \pmod{p}$ with $1 \leq i-j < d$, contradiction!) Since each integer $a, a^2, \dots, a^d$ is a solution to the congruence $x^d - 1 \equiv 0 \pmod{p}$, Proposition 5.8 implies

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that the incongruent solutions modulo p of the congruence $x^d \equiv 1 \pmod{p}$ are given precisely by $a, a^2, \dots, a^d$. Exactly $\phi(d)$ of these d integers have order d modulo p by Corollary 5.5. Thus $f(d) = \phi(d)$ (which certainly implies that $f(d) \leq \phi(d)$), and so the proof is complete. 

Corollary 5.10: Let p be a prime number. Then there are exactly $\phi(p-1)$ incongruent primitive roots modulo p . In particular, primitive roots exist for all prime numbers.

Proof: A primitive root modulo p is an integer of order $\phi(p) = p-1$. There are exactly $\phi(p-1)$ such incongruent integers by Theorem 5.9. 

• Theorem 5.9 asserts the existence of integers

having certain orders. However, the Theorem (4) does not construct these integers.

Example: • Let $p=7$. Theorem 5.9 implies that there exist integers of orders 1, 2, 3 and 6. (these are the divisors of $\phi(7)=6$). Furthermore,

order	# of residue classes	residue classes
1	$\phi(1) = 1$	1
2	$\phi(2) = 1$	6
3	$\phi(3) = 2$	2, 4
6	$\phi(6) = 2$	3, 5

Exercise: Construct the analogous table for $p=13$

Example: Find all incongruent integers having order 6 modulo 19.

Theorem 5.9 tells us that there will be $\phi(6)=2$ such integers. We want to avoid an exhaustive search through the integers from 1 to 18. Lets

first prove that 2 is a primitive root modulo (5)
modulo 19. By Proposition 5.1 we need only
test 2^a where $a \mid \phi(19) = 18$. Thus $a = 1, 2, 3, 6, 9, 18$.

$$2^1 \equiv 2 \pmod{19}, 2^2 \equiv 4 \pmod{19}, 2^3 \equiv 8 \pmod{19},$$

$$2^6 \equiv 7 \pmod{19} \text{ and } 2^9 \equiv 18 \pmod{19}. \text{ Thus}$$

$$\text{ord}_{19} 2 = 18. \text{ Proposition 5.3 tells us that}$$

the first 18 positive integral powers of 2
will generate all reduced residue classes

modulo 19. Proposition 5.4 allows us to

compute the orders of these integral powers of

2. We have

$$\text{ord}_{19}(2^i) = \frac{\text{ord}_{19} 2}{(\text{ord}_{19} 2, i)} = \frac{18}{(18, i)}.$$

We want $\text{ord}_{19}(2^i) = 6$, which happens if and
only if $(18, i) = 3$. Thus $i = 3$ or $i = 15$. We

have

$$2^3 \equiv 8 \pmod{19},$$

and

$$\begin{aligned}
 2^{15} &\equiv (2^4)^3 2^3 \equiv (-3)^3 2^3 \equiv (-27)(8) \\
 &\equiv (-8)(8) \\
 &\equiv 12 \pmod{19}.
 \end{aligned}
 \tag{6}$$

Thus 8 and 12 are the desired incongruent integers having order 6 modulo 19.

- Consult Table 5.1 for the least primitive roots modulo p for some small primes.

The frequency of 2 as the least positive primitive root modulo p suggests the following conjecture.

Conjecture: There are infinitely many primes p for which 2 is the least positive primitive root modulo p .

Conjecture (Artin): If r is a non-square integer other than -1 , then there are infinitely many primes p for which r is a primitive root modulo p .

- Artin's Conjecture is still open! Heath-Brown proved in 1986 that there are at most two integers r for which Artin's conjecture fails!

The Primitive Root Theorem:

- Over the course of the next few lectures we will characterise the integers that have a primitive root.
- The next two Propositions will limit the integers that we need to consider.

Proposition 5.11: There are no primitive roots modulo 2^n where $n \in \mathbb{Z}$ and $n \geq 3$.

Proof: Note that a primitive root modulo 2^n must be odd and have order $\phi(2^n) = 2^n - 2^{n-1} = 2^{n-1}$. Let a be any odd integer. To prove that no primitive root modulo 2^n exists, it suffices to show that

$$a^{2^{n-2}} \equiv 1 \pmod{2^n}.$$

We use induction on n . If $n=3$, we must (8) show that $a^2 \equiv 1 \pmod{8}$. We must have $a \equiv 1, 3, 5, 7 \pmod{8}$, and in all cases we have that $a^2 \equiv 1 \pmod{8}$. Assume that $k \geq 3$, and that the desired result holds for $n=k$ i.e. $a^{2^{k-2}} \equiv 1 \pmod{2^k}$. We must show the desired result holds for $n=k+1$. By the induction hypothesis we have $a^{2^{k-2}} - 1 = 2^k b$ for some $b \in \mathbb{Z}$. Then $a^{2^{k-2}} = 2^k b + 1$, and squaring both sides of the equation gives:

$$\begin{aligned} a^{2^{k-1}} &= (2^k b + 1)^2 \\ &\equiv 2^{k+1} (2^{k-1} b^2 + b) + 1 \\ &\equiv 1 \pmod{2^{k+1}}, \end{aligned}$$

as required.



Proposition 5.12: There are no primitive 9
roots modulo mn where $m, n \in \mathbb{Z}$, $m, n > 2$,
and $(m, n) = 1$.

Proof: We will prove that the order of any
integer modulo mn is less than or equal to
 $\phi(mn)/2$; Since a primitive root modulo mn
must have order $\phi(mn)$, this suffices to prove
the Proposition. Let $a \in \mathbb{Z}$ with $(a, mn) = 1$.
Then $(a, m) = 1$ and $(a, n) = 1$. Put $d := (\phi(m), \phi(n))$
and $k = [\phi(m), \phi(n)]$. Since $\phi(m)$ and $\phi(n)$
are both even we have $d \geq 2$. Thus

$$k = \frac{\phi(m)\phi(n)}{d} = \frac{\phi(mn)}{d} \leq \frac{\phi(mn)}{2}.$$

By Euler's Theorem, we have that
 $a^{\phi(m)} \equiv 1 \pmod{m}$. Taking the $\frac{\phi(n)}{d}$ th power
of both sides of the congruence gives
 $a^k = a^{\phi(m)\phi(n)/d} = (a^{\phi(m)})^{\phi(n)/d}$

$\equiv 1 \pmod{d} \equiv 1 \pmod{m}$. Similarly, (10)
 $a^k \equiv 1 \pmod{n}$. Since $(m, n) = 1$, we have
that $a^k \equiv 1 \pmod{mn}$.

Thus the order of any integer modulo mn
does not exceed $k \leq \frac{\phi(mn)}{2}$, as required.

□

Corollary 5.13: Let $n \in \mathbb{Z}$ with $n > 0$. If
a primitive root modulo n exists, then
 n must be equal to $1, 2, 4, p^m$ or $2p^m$ where
 p is an odd prime and m a positive
integer.

Proof: Left as an exercise.